Techniques for automated planning of access networks

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This paper discusses the application of versatile optimisation algorithms to the process of planning access networks using optical fibre or copper. A cost-effective fibre access solution must make efficient use of the existing network infrastructure. An automated system therefore requires reliable data on the physical network in a suitable form. This has implications for the balance between the planner's expertise and any software tool used in the planning process. Appropriate use of software tools will have a profound impact on the efficiency and working practices of the planning process.

1. Introduction

If customers are to benefit from new broadband services, the access network must be able to handle much higher bit rates. This demands the introduction of new technology, whether based on copper, optical fibre or radio. The challenge lies in devising a cost-effective strategy for modernising BT's access network. Efficient processes for the detailed planning of networks are an essential element in meeting this challenge.

A major asset of an established telecommunications operator is its physical infrastructure — primarily the network of cable routes. The cost of laying new duct is high and it is essential to make best use of what already exists. This leads to a requirement for accurate infrastructure records.

This paper discusses some of the issues associated with incorporating automated elements into the access planning process. A central concern is the availability of reliable network infrastructure data in a form accessible by computer. Some processes to aid this are suggested. Next, an efficient algorithm based on simulated annealing is described which can deal with the large number of variables and constraints typically required in a planning problem. Finally, two realistic examples of optical fibre planning exercises are presented, one consisting of point-to-point deployment of fibre from an exchange and the second of the design of a passive optical network (PON).

2. Network data

The access planning process must yield a specification of the equipment to be deployed, where it is to be installed and how it is to be integrated with the existing network. There are a number of constraints on the planner, including the location and availability of existing plant (e.g. duct space, network access points), restrictions on the location of new plant, the location and level of customer demand, as well as performance-related factors. Automated optimisation algorithms can handle such problems efficiently, but the quality of the solution depends on the quality of the starting data, particularly plant records. In a traditional manual planning environment, geographical plant records are held as pictures (either on paper or as raster scanned images on a computer), with supplementary notes giving more detailed information. Data in this form is not well suited to manipulation by computer. A database is required that allows computer access to all relevant plant information including location.

3. Geographical information systems

The requirement for handling data that has a geographical content is common to a wide range of business es — retail, utilities, distribution, emergency services, etc. This has led to the development of specialised database software — geographical information systems (GIS) — which are now available as commodity software at reasonable cost on a variety of platforms (e.g. PC, Mac, Unix). A GIS adds location information to a database, allowing geographical relationships to be analysed and providing a graphical interface. This makes visualisation and analysis of large and complex geographical datasets possible. As an example, the plant records for an exchange area can be viewed on a map, possibly overlaid on a raster scanned street plan or Ordnance Survey map. Information about individual duct routes or nodes can then be obtained with a mouse click. Records can be amended, proposed new plant can be added, correlation with customer demand can be performed — all in a straightforward way.

The use of a GIS is central to an automated planning process. It provides the interface between the planner and a set of computer-based tools. While automatic planning algorithms can perform complex network optimisations, there are important areas in which the experience and intelligence of a planner are vital — and for which visualisation of the network structure is required. These include validation of existing plant records and decisions on where new duct routes and network access points might be sited. The planning process will then involve a closely integrated mixture of human skills and automated tools.

4. Database maintenance

The requirement for reliable records of the existing network infrastructure to be available to all planners (local and national) indicates the need for a central electronic plant database. However, a distributed approach to data management is preferable. Efficient processes for database maintenance are vital. Changes are frequent, as a result of new installation and maintenance of the existing network. People working in an area, whether in planning or maintenance, are responsible for most of the changes in that area. They also rely on accurate local data to work efficiently which gives an incentive to ensure changes are captured accurately.

Local data can then be aggregated to give a wider picture and made available to any interested user.

As a tool for reliably capturing geographical information, global positioning systems (GPS) may be used, particularly outside urban areas where precise location on a map may be difficult. GPS receives broadcast signals from a number of satellites and uses the information to calculate location typically to within a few metres. GPS systems are now inexpensive and compact, making it feasible for an engineer in the field to automatically capture accurate information about the location of network elements and to incorporate it into the database.

5. Automation in the planning process

It is unlikely that the planning process can be automated to the extent that there is no longer a need for skilled planners. The automated elements of the planning system will assist the planner in various aspects of the planning process.

Figure 1 shows a possible division of the process. The starting point is the infrastructure database. The planner is responsible for validating the data and identifying options for the siting of new plant. This is best left to the planner. The constraints on where to run a new duct, for example, may involve a number of factors requiring detailed local knowledge (e.g. location of roads, ownership of land, wayleaves) which are difficult to include in a general automated tool. Planning a network on a green-field site requires the planner to specify a full range of options for the location of new plant, with associated costs, based on marketing information as well as the details of the site. This might seem to represent a major task, but not all the infrastructure proposed at this stage need be used in the final design. Planners have just to identify any duct route or network access point that they judge might be useful. Exploring the wide range of potential network designs within these limits is a task best automated.

A typical exchange area consists of several thousand duct sections and network nodes, e.g. flexibility points, distribution points (DPs), cable joints. The first step in the planning process is to analyse the input data. This includes identifying the data relevant to the planning problem and resolving any obvious errors. The existing duct network is based on a tree-and-branch structure radiating from the local exchange. All customers must have a physical connection back to the exchange. Isolated sections of network in the plant records can be identified automatically and highlighted on a map. The planner can then correct the data after a physical survey of part of the network, if required.

An area in which the optimisation algorithms described below differ significantly from the manual processes they can replace is their ability to include complex constraints. Planners work to a set of planning rules which ensure that systems operate within technology limits, e.g. power budgets, wavelength dispersion limits, system capacity. However, the rules also impose restrictions so that the manual planning process consistently yields sensible designs in a reasonable time. Working directly with just the

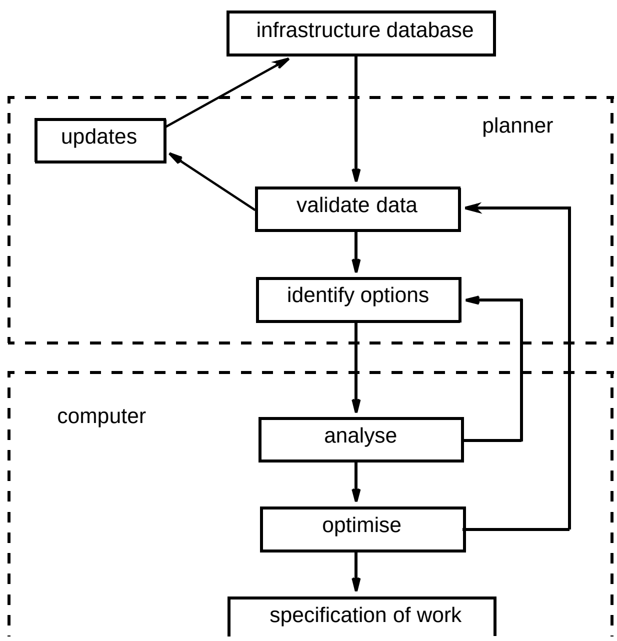

Fig 1 A possible division of the planning process into manual and automated tasks.

technological constraints, the optimisation algorithm may reach a better solution. A further advantage of automated network design is that a detailed breakdown of the plant deployed at any point in the network can be obtained. Workplans can be generated for the entire optimised network, or for any subset of it if construction is to be phased. Automation of this activity can yield valuable time savings. The final stage in the planning process is to verify that the network as designed can be constructed. This involves, for example, a partial physical survey of the network infrastructure to verify that the data used is accurate. If there are significant errors the data can be corrected and the network replanned in the light of the fresh information.

6. Simulated annealing

Any optimisation problem may be regarded as the problem of finding the global maximum or minimum of a function of a number of variables subject to a number of constraints. In some cases it is possible to find analytic solutions but, in general, this is difficult or impractical, particularly for problems with large numbers of variables.

Simulated annealing is a statistically based optimisation technique, with a foundation in statistical physics [1] first developed at IBM [2]. It can be efficiently applied to optimisation problems with a large number of variables and allows constraints to be defined in a flexible way. Location of the global optimum is not guaranteed, but measures are taken to avoid converging on a local optimum close to the starting point.

A set of variables is defined to describe the state of the system and the constraints are incorporated into a measure of how good a particular configuration is. This is a generalised cost or value function which is then optimised. In the examples presented here, the optimisation is driven by a cost function which is minimised.

One of the variables is selected at random and changed. The change in the cost function is then calculated and a decision made as to whether to accept or reject the change. Any change which improves the cost function is accepted, but changes which make it worse are not always rejected. They are accepted with a probability P, given by:

$$P(\Delta) = e^{-\frac{\Delta}{T}} \qquad \Delta > 0$$

where Δ is the change in cost function and T is a parameter analogous to temperature. It is this feature of the optimisation that allows local maxima to be avoided. The temperature parameter starts at a high value so the probability of acceptance approaches unity and is gradually reduced as the optimisation proceeds. At low temperatures, P tends to zero and only changes leading to an improvement in the cost function are accepted. Provided the cooling rate is slow enough, the solution converges to one which is close to the global optimum. The technique clearly has wide application and has, for example, been applied at BT Laboratories to the design of diffractive optical elements [3] and the training of neural networks [4, 5].

In a typical tree-and-branch access network optimisation problem, there might be a few hundred nodes. The variable parameters are the possible routes back to the trunk — of which each node may have several. The cost function will closely reflect the real cost, including contributions from fibre cables, joints, splitters, subduct, digging new duct, cabinets, etc, and will also include other constraints (such as power budget limits, available bandwidth) given some weighting relative to the real costs. Simulated annealing is very well suited to problems of this type with many variables and complex constraints.

Modern desktop PCs are now powerful enough to allow simulated annealing runs of this size and complexity to run in minutes, making it realistic to use the technique interactively within the planning process.

7. Examples

To illustrate the application of automated techniques to realistic planning applications, two examples have been considered. The first involves point-to-point provision of fibre from exchanges to primary connection points (PCPs) in a fairly large area (three exchange areas aggregated). This could form part of a possible strategy for deployment of a broadband network. Optical fibre is brought closer to customers than at present and the final connection from PCP to customer can then be broadband over copper (at high bandwidth because the length of copper is short), or eventually fibre.

The second example is the design of a PON. This gives shared fibre access to a relatively small number of customers (~30), served from a nearby node at which fibre is already available (say a PCP). MapInfo ®, a commercial GIS package, with MapBasic ® [6] have been used to develop an interface to these optimisation programs.

The examples use plant data from the Bristol area. While the geographical locations of cable routes and plant used in the examples accurately reflect the plant records, other information (materials and labour costs, duct occupancy, etc) has been estimated. Therefore, the detailed fibre plans are only intended to illustrate the planning process.

8. Point-to-point fibre planning

Three adjoining exchange areas in Bristol (Bedminster, Bishopsworth and Whitchurch) were selected. The planning problem was to provide point-to-point fibre connection from each of two head nodes (chosen arbitrarily) to all the PCPs in the combined area, using the existing network infrastructure.

Feeding fibre from two head nodes is a way of achieving some measure of network resilience. The high bandwidth of optical fibre means that a single network failure can result in loss of service for a large number of customers. For example, in the scenario considered here, there are about 50 000 customer connections. Feeding fibre from a single point would mean that failure at, or close to, that point could affect a large proportion of those customers. Using a second head limits the impact of a single network failure to a fraction of the customers. It does not offer as much resilience as full route separateness, but may offer a cost-effective means to limit the impact of network failure for those customers not requiring guaranteed availability of service.

The area under consideration contains about 12 000 cable route sections (either duct, direct buried or aerial cable routes) and a similar number of nodes, e.g. PCPs, DPs, poles. The basic topology of the network is a tree-and-branch structure from the exchange within each of the exchange areas, to PCPs, and then to customers.

Most of the infrastructure is irrelevant to the problem of interconnecting PCPs and the exchange. Branches connecting DPs back to PCPs, for instance, can be ignored. The first step is to analyse the structure of the sub-network interconnecting the PCPs and head nodes. This reduces the size of the problem to about 1000 cable route sections — making the optimisation more manageable — and checks the network connectivity. The sub-network connecting all the PCPs and the two head nodes is shown in Fig 2.

The planner can then view the simplified network structure and investigate any unconnected regions. In addition, the planner may judge that connectivity could be improved by the addition of some extra cable routes. For instance, the access network has historically been organised by exchange area. The size of each exchange area is convenient for conventional copper transmission and switching systems, based on the number of customers to be served. Typically, there are few cable routes in the access network which cross exchange boundaries. The increased capacity of optical fibre systems means that larger areas may be more appropriate — concentrating expensive equipment in fewer locations. If exchanges are bypassed, routes crossing exchange boundaries become particularly important and their relative scarcity may mean that they become bottlenecks. However, cable routes often run close to the boundary on each side. Addition of a small quantity of new duct could, in these cases, dramatically improve the connectivity and the final network design.

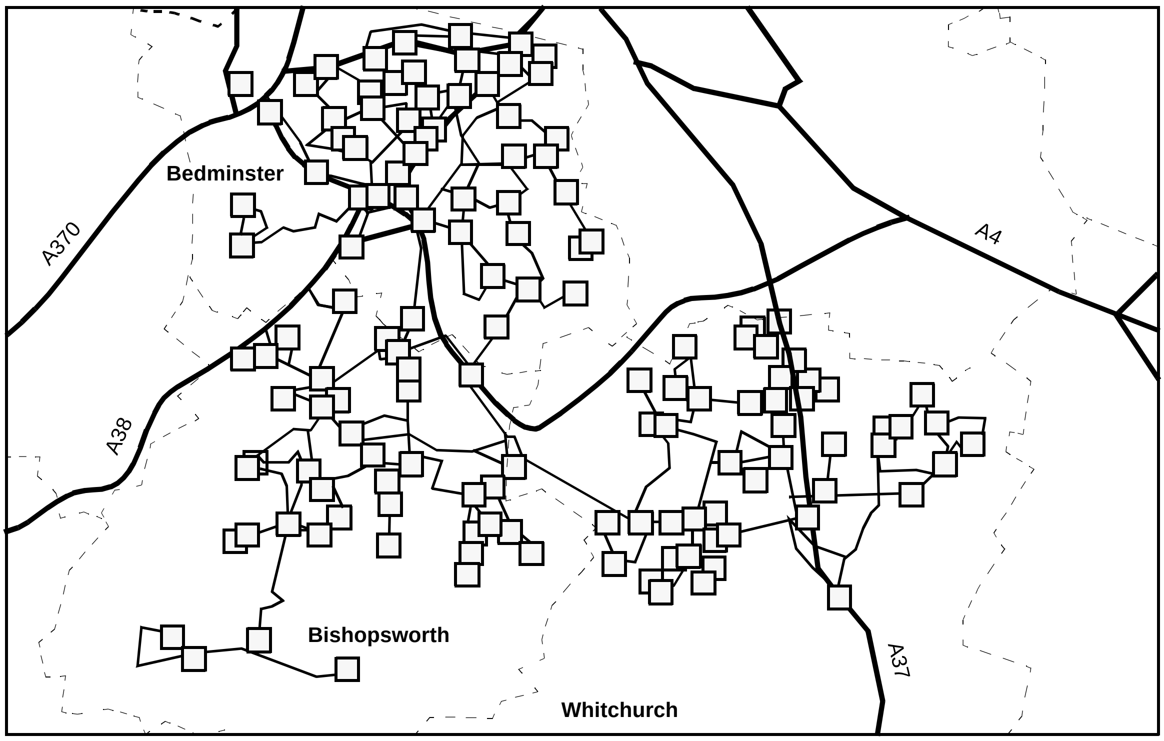

Fig 2 Map of the three-exchange area showing only the simplified sub-network connecting the PCPs.

New duct sections can be proposed by the planner in suitable locations, and build costs associated with them. The planner can be fairly uninhibited about proposing new routes in this way. The optimiser will include them in the problem — with the appropriate cost — and building of the new routes is only necessary if they are required in the best design. As the computer time required for a problem of this size is modest (about a minute, typically), repeated analysis of the network and amendment by the planner is feasible.

Once a satisfactory description of the duct network has been generated, optimisation of the network design is carried out using a simulated annealing algorithm.

For this example, this takes a few minutes on a desktop PC. The optimised solution is shown in Fig 3.

The final design specifies the number of fibre cables in each duct section, indicated by the thickness of the lines in Fig 3, and it is straightforward to extract a detailed inventory of materials and labour required for constructing either the whole network or any portion of it.

9. PON design

Few customers will require the transmission capacity of a direct fibre link to their premises. A PON is a way of sharing the capacity of a fibre among a group of customers using passive fibre splitters. Typically, up to 24 customers are served from a single fibre. The planning problem involves fewer cable routes and nodes than the point-to-point example. However, there is an additional constraint on the solution — the received optical power at the customer's fibre termination must lie within certain limits. This constrains the total loss (fibre transmission loss and splitter loss, principally) between the PON head and the customer nodes. There are also limits to the number of termination points of the optical network served by a single PON, and on the total bandwidth. This problem is difficult to plan manually. In the point-to-point example above, the best solution is close to that which minimises the total fibre length. This is because most of the costs are associated with fibre cable and new duct build — both length dependent. This sort of solution can be visualised quite easily. In the PON case, a significant part of the network cost is associated with the fibre splitters. The best solution will have to make efficient use of these expensive elements and can be difficult to guess at intuitively.

The example chosen here is derived from the area surrounding a single PCP in one of the Bristol exchange areas. There are 34 DPs in the example, each assumed to have a single customer connection and a requirement for 10 × 64 kbit/s channels.

Splitters with any available size of split (e.g. 4-way, 8-way) can be sited at any node. The cost function in this example includes real costs associated with fibre cable, splitters, new duct build, etc. The power budget at the customers is constrained to lie close to 15 dB, representing a 32-way split. There is not a rigid insistence on a 32-way split to each customer, but there is a penalty cost associated

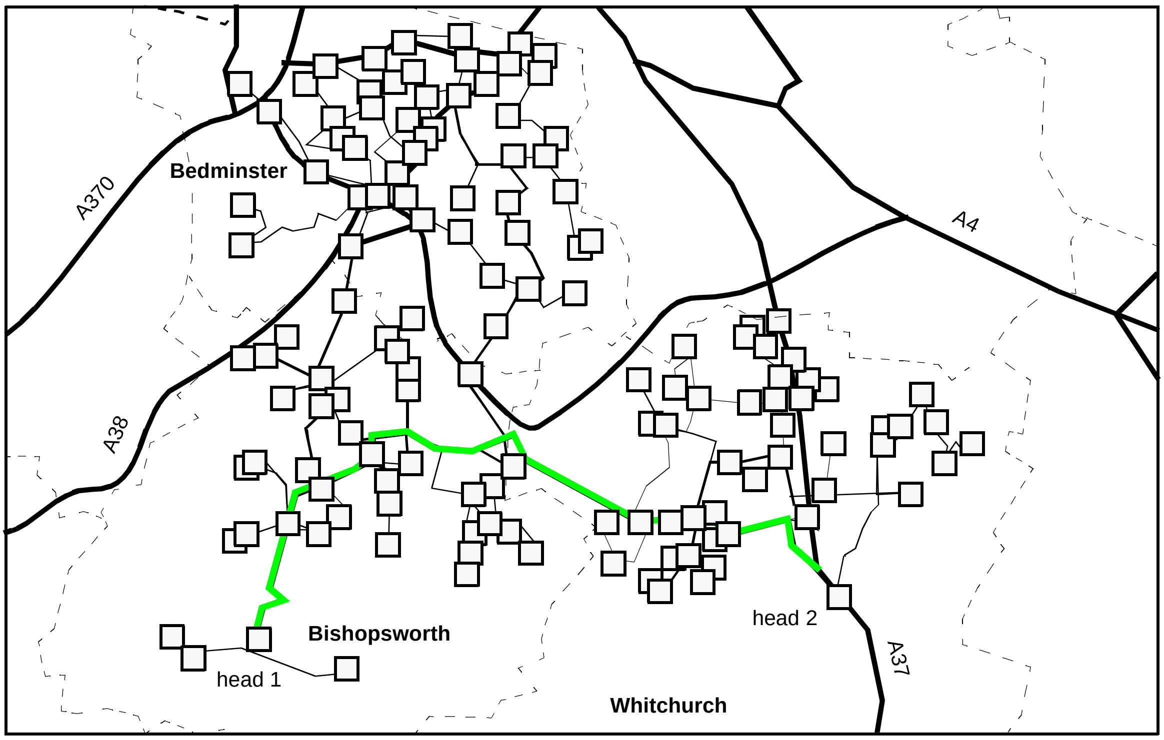

Fig 3 Map of the three-exchange area showing the optimised routeing of cables through the duct network to ensure connection of each PCP to both of the two head nodes. The thickness of the lines represents the number of fibres within each duct section.

with deviation from the target value. The number of network terminations and the total number of channels on a single PON are also constrained.

The final design specifies the location and size of splitters and fibre cables and is shown in Fig 4. There are two separate PONs, with 14 and 20 customers respectively, all originating from the node marked as PON head. The thickness of the lines indicates the number of fibres in each section of duct. Each splitter site houses at least one splitter. The design shown uses nine 4-way and two 8-way splitters.

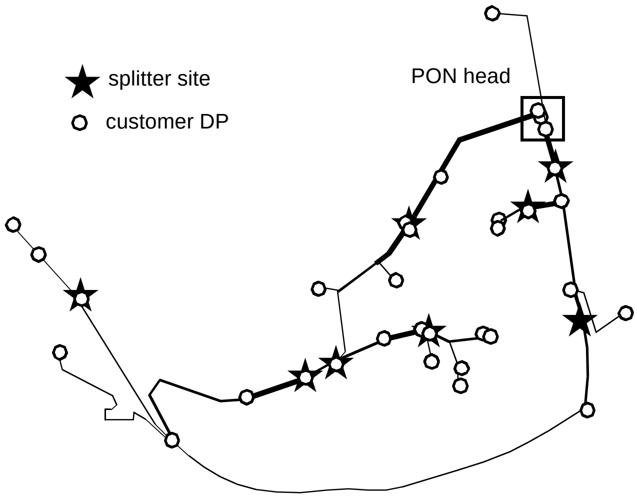

Fig 4 Map of the optimised PON design to serve 34 customers (o) from the PON head node. Splitter sites (which may house more than one splitter) are indicated by stars (★). Two PONs are overlaid to provide the required service.

10. Conclusion

Advances in flexible optimisation techniques and readily available computing power now mean that there is the potential for the process of planning telecommunications networks to be greatly improved. The introduction of automated techniques into the network planning process will change the role of the human planner.

It is now feasible to develop automated tools both for the analysis and visualisation of network data and for the optimisation of network design, making best use of the existing network infrastructure.

A prerequisite to the use of automated network planning tools is the availability of reliable data on the physical network in vector electronic form, suitable for manipulation by computer. Any database of the scale of BT's physical network will contain significant errors and it must be the responsibility of the planner to ensure appropriate use of the automated elements of the planning process. The offering of efficient automated planning tools to a skilled and experienced planner will result in significant benefits in the planning process.

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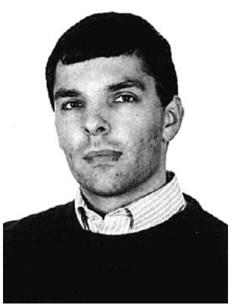
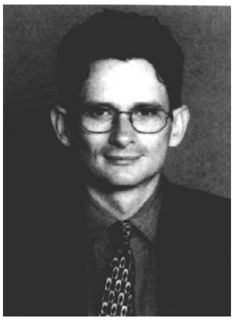
6 ManInfo and MapBasic are registered trademarks of MapInfo Corporation, One Global View, Troy, New York, USA.



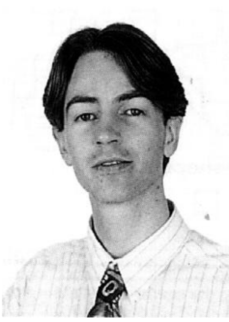
After receiving a BA in Natural Sciences from Cambridge University and a PhD in Physics from University of Surrey, Mike Fisher joined BT Laboratories in 1988. He worked on nonlinear optics in semiconductor materials and on the design, fabrication and characterisation of novel optoelectronic devices.

He then moved into the networks research unit where he is applying computer techniques to network design processes.



Tim Rea obtained a degree in Glass Science and Technology from the University of Sheffield in 1993 and then started a PhD before joining BT Laboratories in 1995 to work on fluoride glasses.

He is now involved in work related to future access and is currently focusing on the use of Global Positioning Systems and other new technology in the access network.



Andrew Swanton received a BSc degree in Applied Physics from Bath University in 1993 and an MSc degree in the Physics of Laser Communications from Essex University in 1994. He joined BT in 1994 where he has been involved in the design and fabrication of optical phase masks using high resolution lithography.

He is now involved in future access technology where he has been de-veloping computer interfaces for visualising and manipulating electronic access plant databases.

Mark Wilkinson received a BA degree in Natural Sciences (Physics) from Cambridge University in 1989.

After graduating he joined BT Laboratories in the optical physics division where he began research on optical waveguide devices. This included work on fibre sensors and a computer design technique for waveguide grating elements.

In 1994 he joined the networks research unit to work on network modelling. He is currently working with geographical data for use in automated planning tools and is also involved in network development for advanced videoconferencing.

He is a Member of the Institute of Physics.

David Wood received a BA (first class honours) and PhD in Mathematics from Cambridge University. He joined BT Laboratories in 1984, after academic research work in dynamical systems and nonlinear waves. His initial work at BT was on the nonlinear physics of solitons and laser mode-locking. He then moved into the area of diffractive optics and set up a team to design and make holographic components for optical interconnections and antennas. Next, he moved his research focus again, to apply these modern computer design techniques to networks — both for core, switching networks and fibre deployment in local access. He set up and led this project team, to ensure the practical take-up within BT of automated techniques for fibre-access planning, and other modern processes of data acquisition and fault location. He is now Technical Director of ‘The Network Design House’.